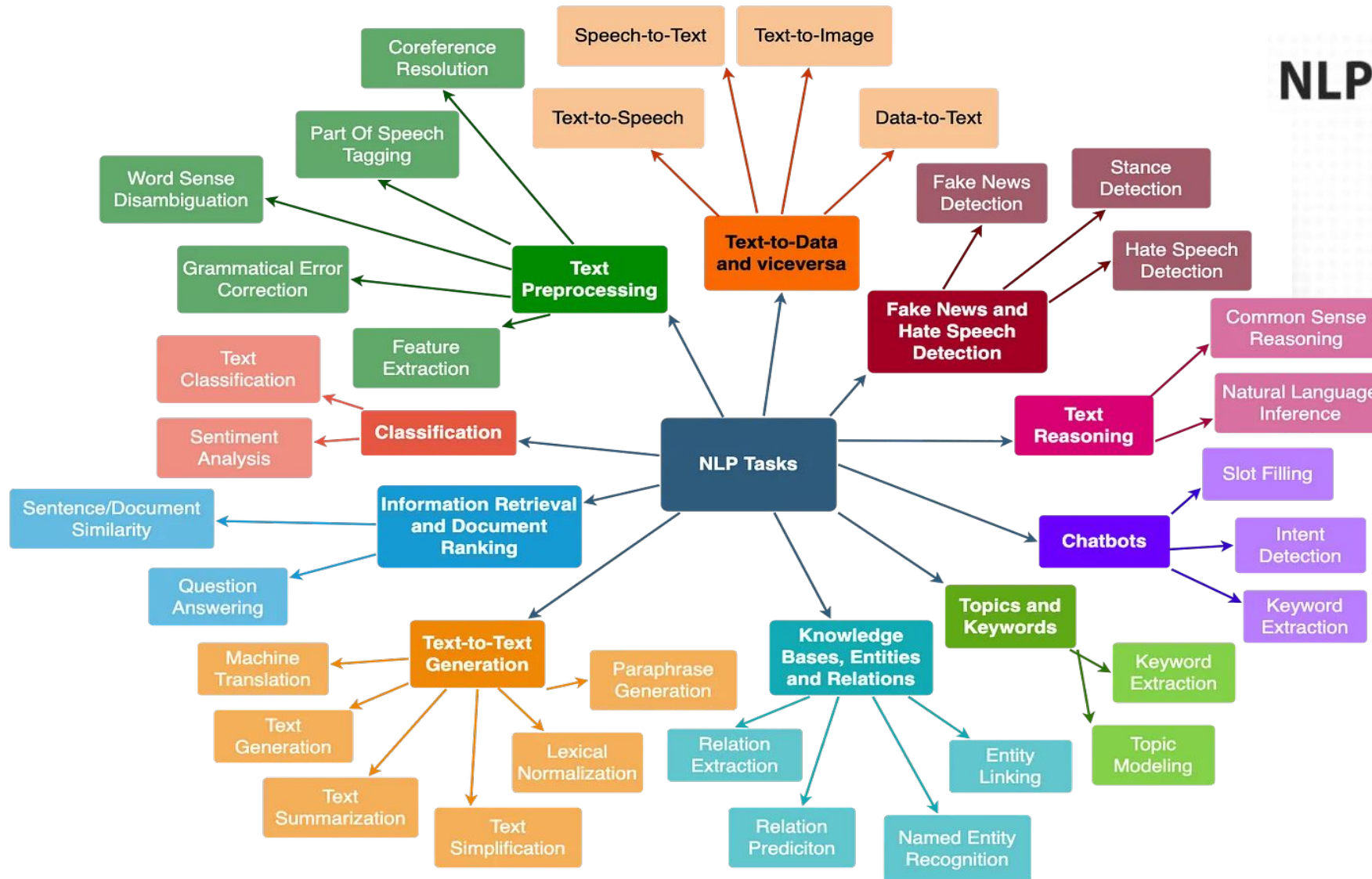




NLP tasks around us:

- suggest in search
- automatic gmail replies
- machine translation

Chatbot

Hey, how are you?

Busy as usual, but fine :)

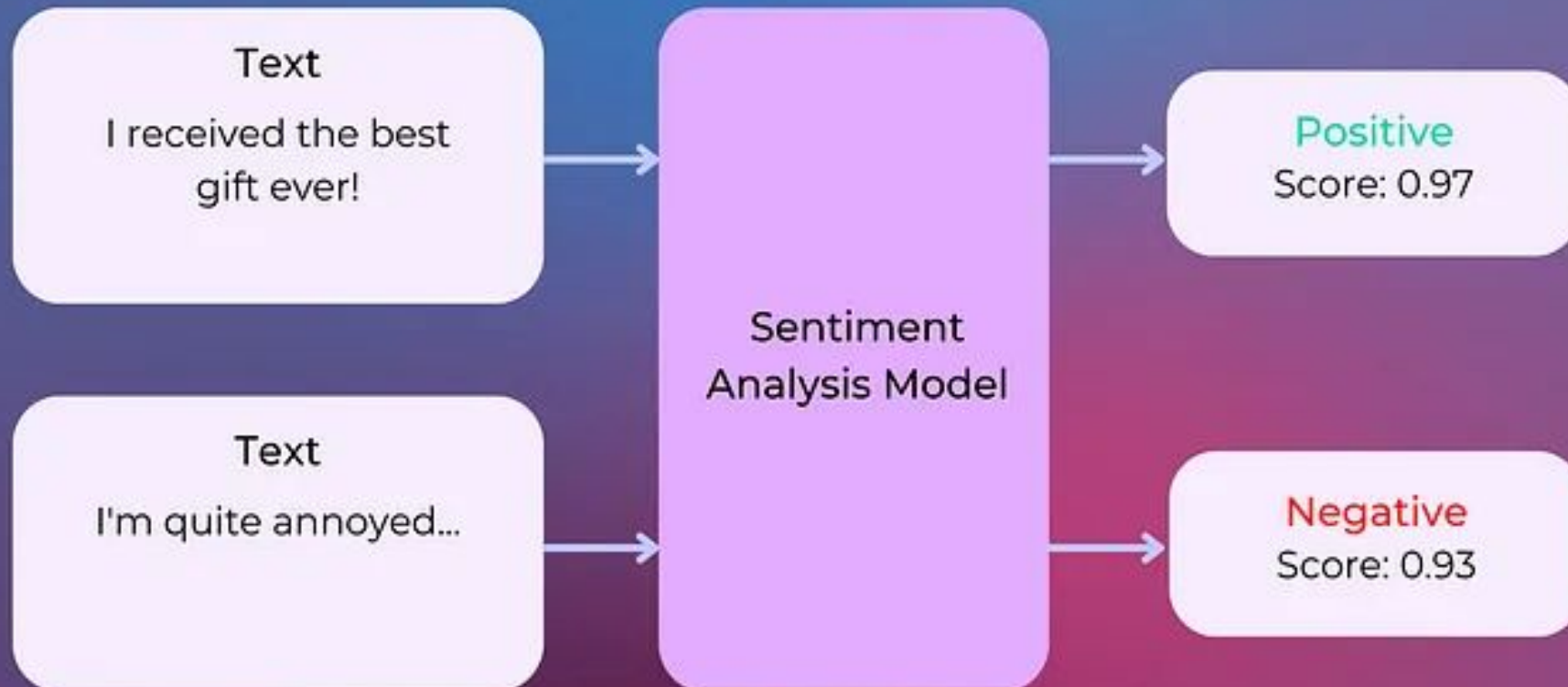
How can I help you?

I am looking for a cheap trip for the holidays...

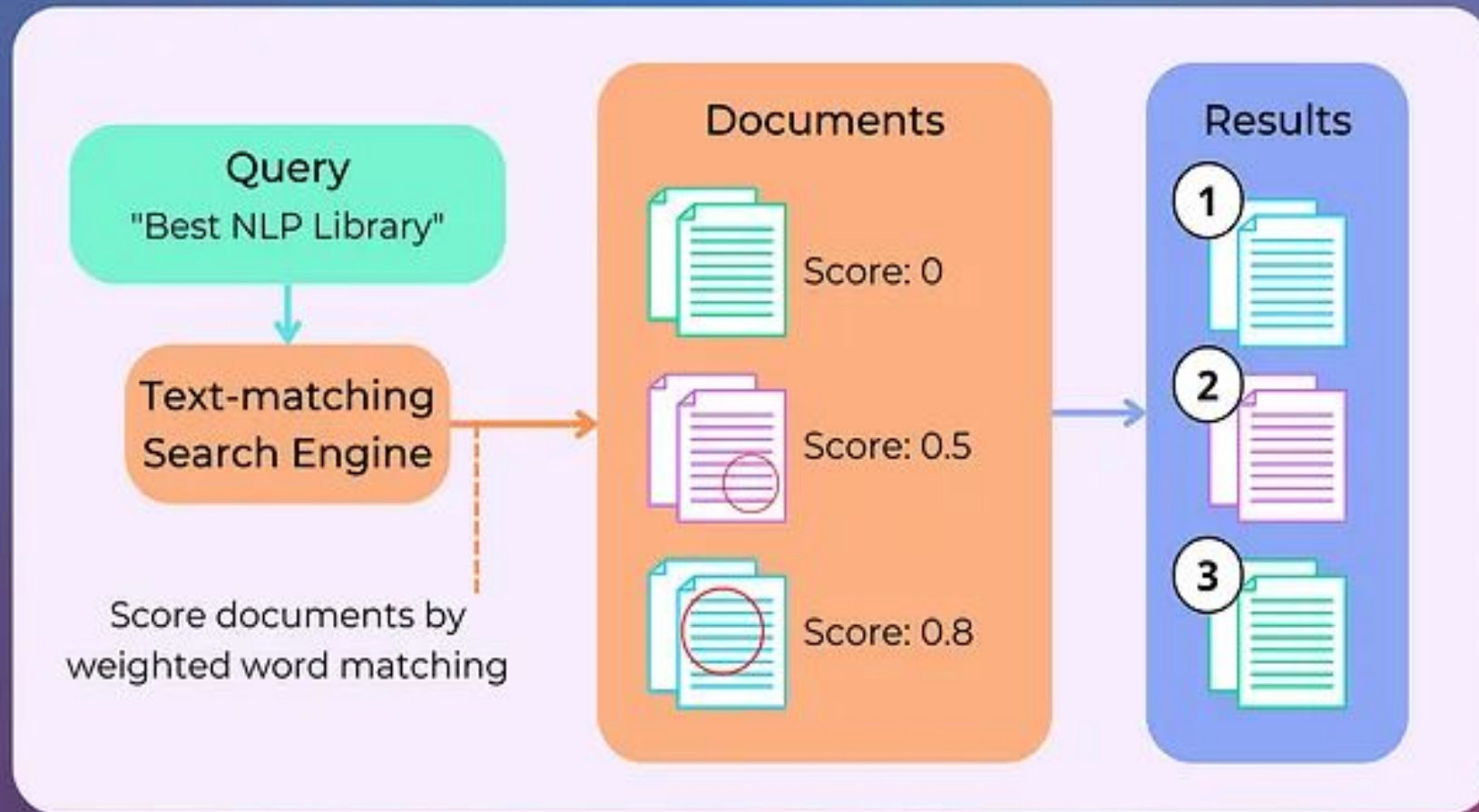
Do you want to relax on the beach?

Oh no please

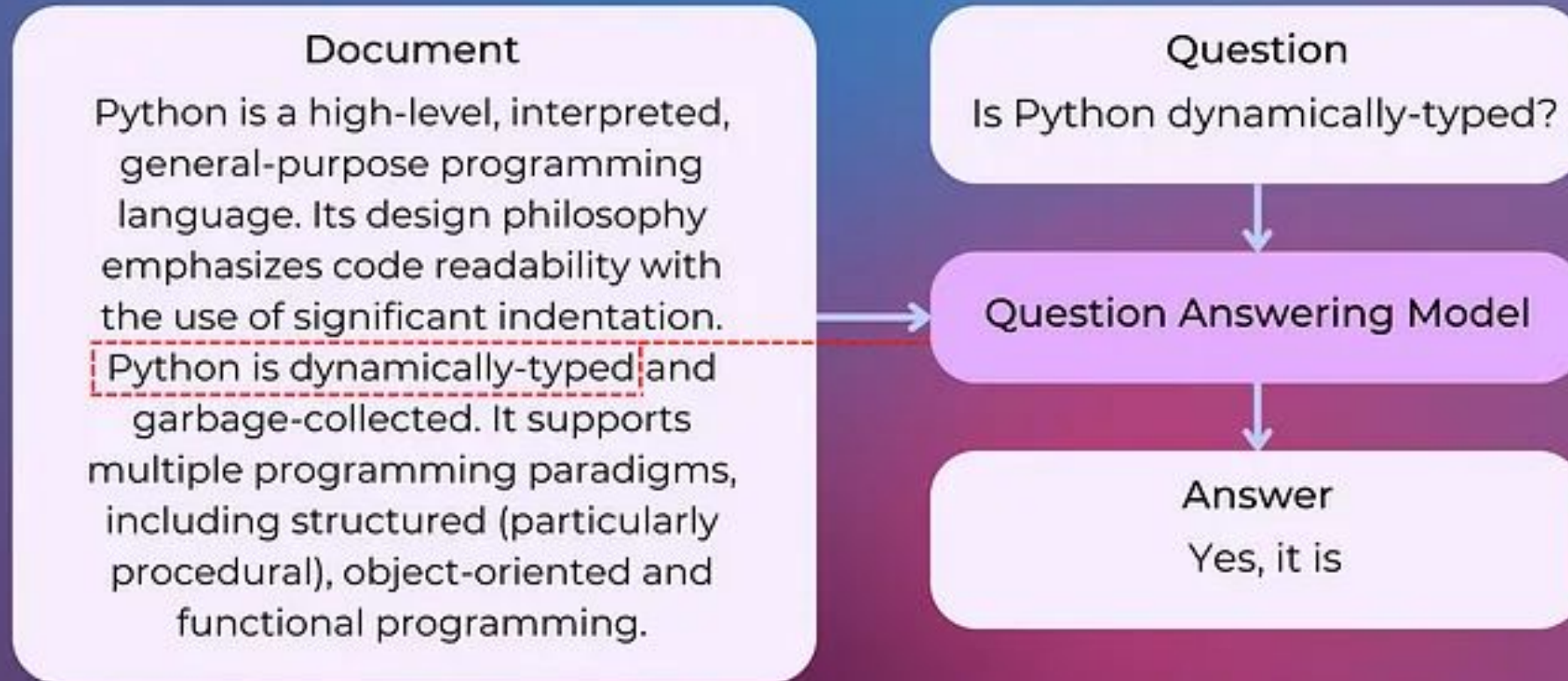
Text Classification - Sentiment Analysis



Search Engine - Text Matching



Generative Question Answering



Named-Entity Recognition

Space Exploration Technologies Corp. **ORG** (doing business as SpaceX) is an **American NORP** spacecraft manufacturer, space launch provider, and a satellite communications corporation headquartered in **Hawthorne GPE**, **California GPE**. It was founded in **2002 DATE** by **Elon Musk PERSON**, with the goal of reducing space transportation costs to enable the colonization of **Mars LOC**.

Building a Knowledge Graph from Texts

Amazon, Microsoft and Google face UK probe over dominance in cloud computing

CNBC - 5 hours ago



Google Surveys Is Shutting Down; Here Are 6 Alternatives

Search Engine Journal - 17 hours ago



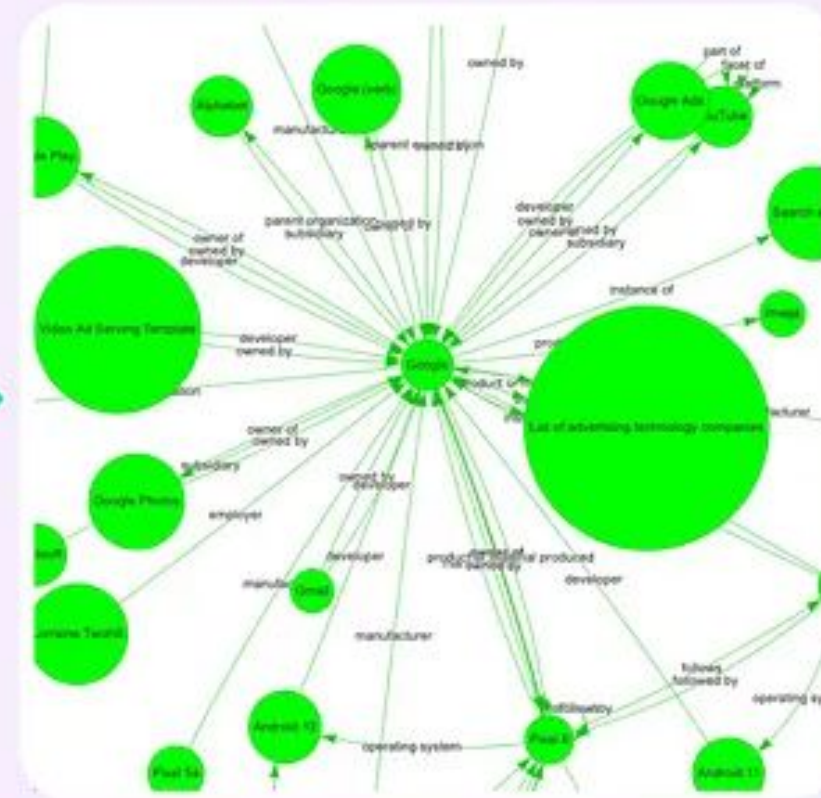
Google's new Chromecast costs \$30 — and it has a remote

TechCrunch - 21 minutes ago



More Google Ad incidents and issues

Search Engine Land - 17 hours ago



Archive Connected Papers - NLP Knowledge Graph Based Innovation



Text Generation

Input

Natural Language Processing (NLP) is a subfield of artificial intelligence that is concerned with the interactions between computers and human (natural) languages.

Text Generation Model

Output

It deals with the automatic manipulation of natural language, such as speech recognition, natural language understanding, and natural language generation. NLP is a relatively new field, having only emerged in the 1950s. However, it has grown rapidly in recent years, with advances in machine learning and artificial intelligence. NLP is used in a variety of applications, such as automatic summarization, question answering, machine translation, and text-to-speech synthesis.

Corpus

- *Definition:* A large and structured set of texts, usually stored electronically used for linguistic research and NLP tasks.
- *Example:* A collection of all articles from Wikipedia.

Documents

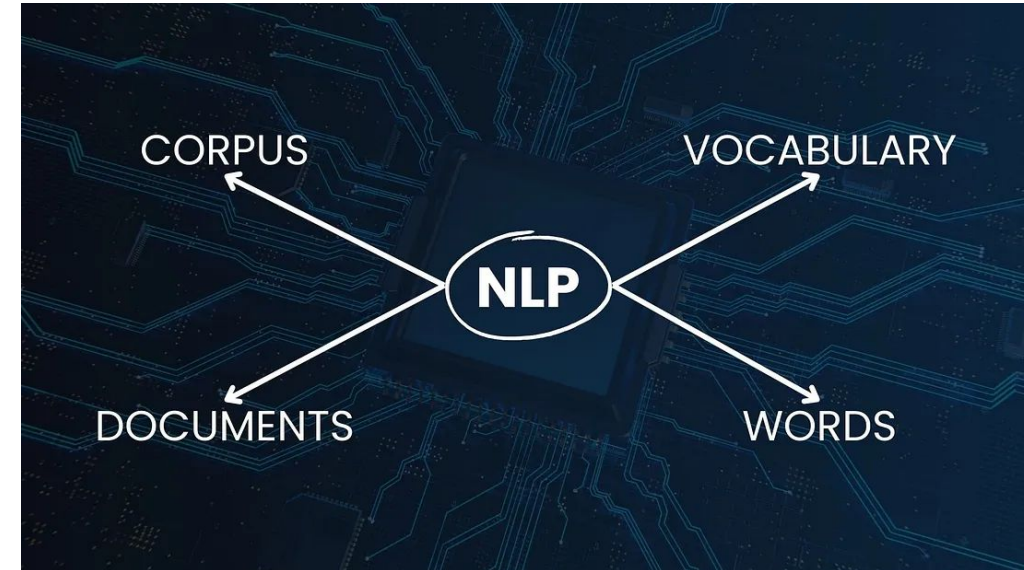
- *Definition:* Individual pieces of text within a corpus.
- *Example:* Each Wikipedia article in the corpus.

Vocabulary

- *Definition:* The set of all unique words used across a corpus.
- *Example:* Words like "computer", "language", "processing" found in the Wikipedia corpus.

Words

- *Definition:* Independent lexical units; the smallest units of meaning in a language.
- *Example:* "Natural", "Language", "Processing", "NLP".



Tokenization is a process that splits an input sequence into so-called tokens

- You can think of a token as a useful unit for semantic processing
- Can be a word, sentence, paragraph, etc.

An example of simple whitespace tokenizer

- `nltk.tokenize.WhitespaceTokenizer`

This is Andrew's text, isn't it?

- Problem: "it" and "it?" are different tokens with same meaning

Let's try to also split by punctuation

- `nltk.tokenize.WordPunctTokenizer`

This is Andrew ' s text , isn ' t it ?

- Problem: "s", "isn", "t" are not very meaningful

Example Tokenization Imp.

```
import nltk  
text = "This is Andrew's text, isn't it?"
```

```
tokenizer = nltk.tokenize.WhitespaceTokenizer()  
tokenizer.tokenize(text)  
  
['This', 'is', "Andrew's", 'text,', "isn't", 'it?']
```

```
tokenizer = nltk.tokenize.WordPunctTokenizer()  
tokenizer.tokenize(text)  
  
['This', 'is', 'Andrew', "'", 's', 'text', ',', 'isn',  
 "'", 't', 'it', '?']
```


We may want the same token for different forms of the word

- wolf, wolves → wolf
- talk, talks → talk

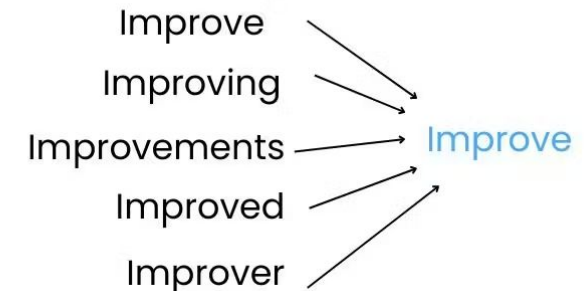
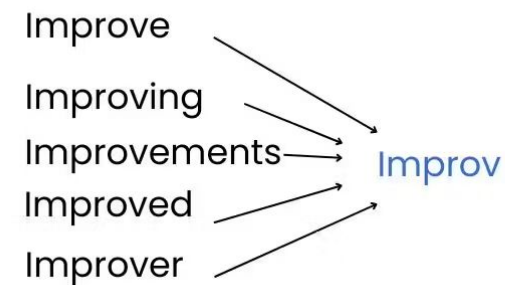
Stemming

- A process of removing and replacing suffixes to get to the root form of the word, which is called the **stem**
- Usually refers to heuristics that chop off suffixes

Lemmatization

- Usually refers to doing things properly with the use of a vocabulary and morphological analysis
- Returns the base or dictionary form of a word, which is known as the **lemma**

Stemming vs Lemmatization



Porter's stemmer

- 5 heuristic phases of word reductions, applied sequentially
- Example of phase 1 rules:

Rule	Example
SS ES → SS	caresses → caress
IES ES → I	ponies → poni
SS → SS	caress → caress
S →	cats → cat

- nltk.stem.PorterStemmer
- Examples:
 - feet → feet cats → cat
 - wolves → wolv talked → talk
- Problem: fails on irregular forms, produces non-words

WordNet lemmatizer

- Uses the WordNet Database to lookup lemmas
- `nltk.stem.WordNetLemmatizer`
- Examples:
 - feet → foot cats → cat
 - wolves → wolf talked → talked
- Problems: not all forms are reduced
- Takeaway: we need to try stemming or lemmatization and choose best for our task

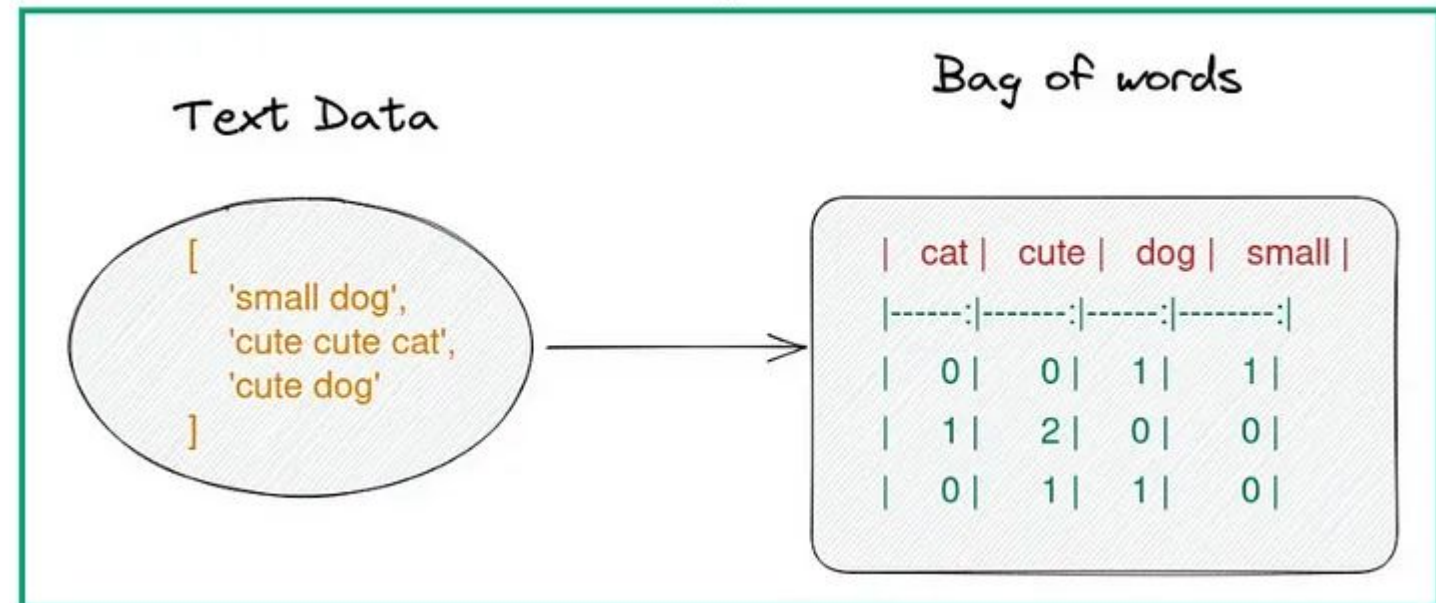
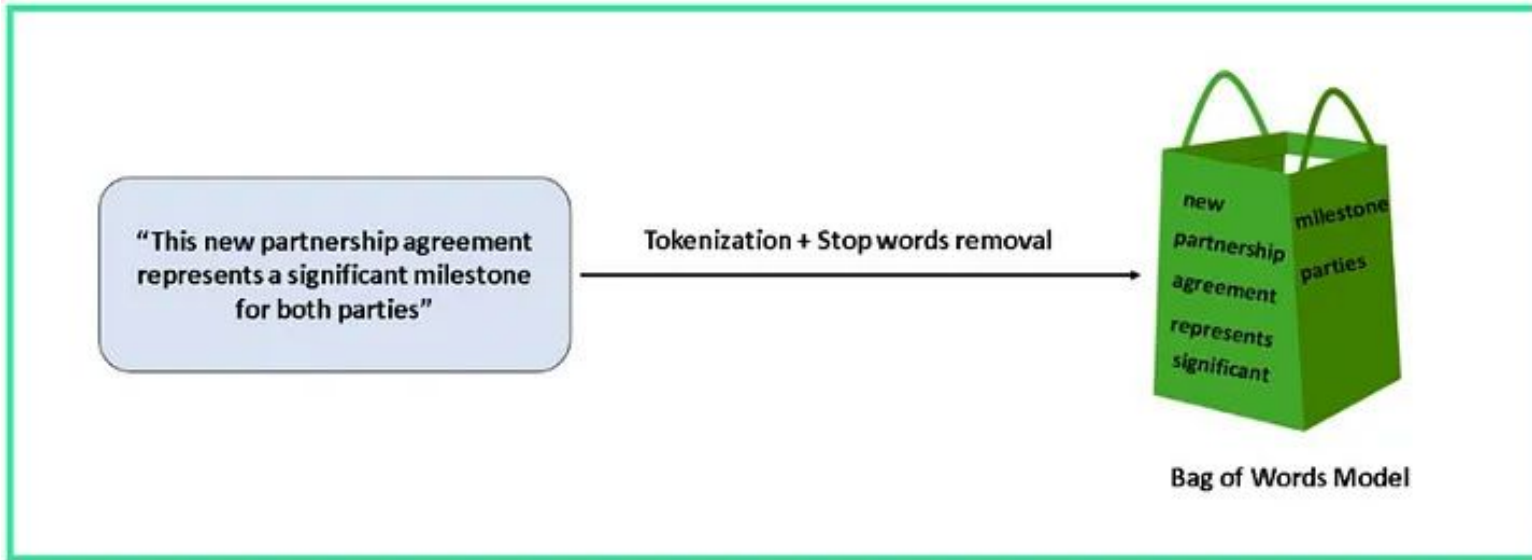
Normalizing capital letters

- Us, us → us (if both are pronoun)
- us, US (could be pronoun and country)
- We can use heuristics:
 - lowercasing the beginning of the sentence
 - lowercasing words in titles
 - leave mid-sentence words as they are

Acronyms

- eta, e.t.a., E.T.A. → E.T.A.
- We can write a bunch of regular expressions → hard

Bag of words



- Problems:

- we lose word order, hence the name “bag of words”
- counters are not normalized

Bag of words

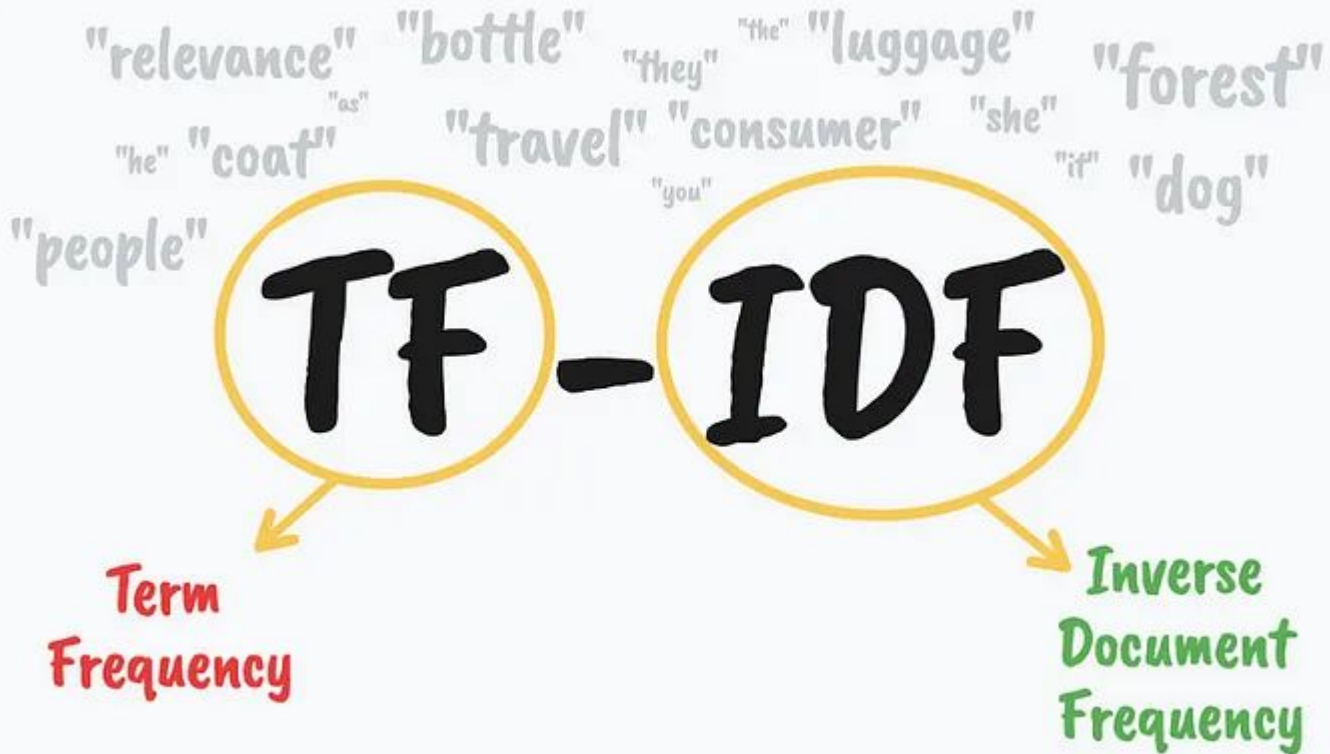
Let's preserve some ordering

Solutions:

We can count token pairs, triplets, etc.

- Also known as n-grams
 - 1-grams for tokens
 - 2-grams for token pairs
 - ...

	good movie	movie	did not	a	...
good movie	1	1	0	0	...
not a good movie	1	1	0	1	...
did not like	0	0	1	0	...



$$TF(t, d) = \frac{\text{number of times } t \text{ appears in } d}{\text{total number of words in } d}$$

$$IDF(t) = \log \frac{\text{Total number of documents}}{\text{number of documents that contain } t}$$

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$